



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING

(25CS1201E)

2025-26, Term-III

Project Problem Statement Submission Form

Sec No- 6	Team No- 21	Project Title- Doctor Appointment Booking System
Team Lead Roll Number		2520030163
Team Lead Name		T. DHARANIDHAR
Team Member Roll Number		2520030325
Team Member Name		ABHIDEV

1. Abstract

The **Doctor Appointment Booking System** is a web-based healthcare application designed to simplify and digitize the process of scheduling medical appointments. In traditional healthcare systems, booking appointments often involves manual procedures, long waiting times, and difficulty in checking doctor availability. This project aims to overcome these challenges by providing an efficient, user-friendly, and centralized online platform for appointment management.

The system allows patients to search doctors based on specialization, location, and availability. Users can view doctor profiles, select preferred appointment dates using an interactive calendar, choose available time slots, and confirm bookings instantly. The application also provides facilities such as appointment reminders, booking history, cancellation, rescheduling, and patient profile management for better convenience and accessibility.

An administrative panel is provided to manage doctor profiles, schedules, patient appointments, and booking records efficiently. Administrators can add new doctors, update availability slots, monitor appointments, and manage overall system operations. This helps improve hospital workflow, reduce scheduling conflicts, and enhance service quality.

The frontend of the application is developed using **React.js** to provide a responsive and interactive user interface, while the backend uses **Node.js and Express.js** for server-side processing and API integration. Data is managed using **MySQL or Local Storage** depending on implementation requirements. The system ensures efficient appointment management, reduces manual workload, improves patient satisfaction, and supports digital transformation in healthcare services.

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2. Module Mapping

Module Name	Assigned Roll Number(s)	Signature
User & Admin Management System	2520030163	
Doctor Discovery Engine	2520030163	
Appointment & Scheduling System	2520030325	
Patient Dashboard & UI Integration	2520030325	

3. Module Details

Authentication: Handles secure user registration and login for patients and administrators..

User & Admin Management System: Handles secure user registration and authentication for both patients and administrators.

- Provides the Admin Panel to add or manage doctor profiles, update schedules, monitor all appointments, and oversee general users.
- Manages the overarching database architecture to securely store user, doctor, and appointment data.

Doctor Discovery Engine: Integrates the search functionality, allowing users to look up doctors by name, availability, and specific criteria.

- Includes the specialty filtering system, enabling patients to narrow down doctors based on medical fields (e.g., Dentist, Cardiologist, Neurologist).

Appointment & Scheduling System: Powers the interactive calendar for users to select dates and view available time slots.

- Manages the core booking logic, giving users the ability to confirm, cancel, or reschedule their existing appointments.
- Handles automated appointment reminders and notifications for upcoming visits.

Patient Dashboard & UI Integration: Manages the patient profile space where personal details and medical preferences are stored.

- Displays the booking history, showing previously booked appointments along with their current status.
- Ensures the entire application maintains a clean, responsive, and user-friendly frontend design.

4. Frameworks / Platforms / Software Used

HTML5, CSS, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL(optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, Local Storage, Bootstrap / Tailwind CSS

Guide Name and Signature (remarks if any) _____



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